INCH-POUND

MIL-C-81659B
AMENDMENT 2
1 October 1990
SUPERSEDING
AMENDMENT 1
12 April 1983

MILITARY SPECIFICATION

CONNECTORS, ELECTRICAL RECTANGULAR, CRIMP CONTACT

This amendment forms a part of Military Specification MIL-C-81659B, dated 2 August 1976, and is mandatory for use by all Departments and Agencies of the Department of Defense.

PAGE 4

Paragraph 2.1 Issues of documents.

STANDARDS

MILITARY

Delete:

MS3156(NAVY)	Tool, Insertion and Removal, Contact, MIL-C-81659 Connector (Series 2)
MS3168(NAVY)	Contacts, Electric, Coaxial, Pin, Crimpable, Size 1
MS3169(NAVY)	Contacts, Electric, Coaxial, Socket, Crimp, Size 1
MS3170(NAVY)	Contacts, Electric, Coaxial, Pin, Crimp, Size 3
MS3171(NAVY)	Contacts, Electric, Coaxial, Socket, Crimp, Size 3
MS3172(NAVY)	Contacts, Electric, Coaxial, Pin, Crimp, Size 5
MS3173(NAVY)	Contacts, Electric, Coaxial, Socket, Crimp, Size 5
MS3174(NAVY)	Contacts, Electric, Coaxial, Pin, Crimp, Size 7
MS3175(NAVY)	Contacts, Electrical, Coaxial, Socket, Crimp, Size 7
MS3176(NAVY)	Contacts, Electrical, Coaxial, Pin, Crimp, Size 9
MS3177(NAVY)	Contacts, Electrical, Coaxial, Socket, Crimp, Size 9

AMSC N/A FSC 5935

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PAGE 4 (continued)

MS3447(NAVY)	Tool, Insert-Extract, Wired Contact, Electric Connector, Sizes 12, 16 and 20
MS27187(NAVY)	Plug, End, Seal, Electrical Connector
MS90455(NAVY)	Tool, Insertion, Contact, Connector
MS90456(NAVY)	Tool, Removal, Contact, Connector

SPECIFICATIONS

MILITARY

Add:

MIL-I-81969/1	Installing and Removal Tools, Connector Electrical Contact, Type III, Class 2, Comp. C
MIL-I-81969/14	Installing and Removal Tools, Connector Electrical Contact, Type III, Class 2, Comp B
MIL-I-81969/17	Installing and Removal Tools, Connector Electrical Contact, Type I, Class I, Comp. C
MIL-I-81969/19	Installing and Removal Tools, Connector Electrical Contact, Type II, Class I, Comp C

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Paragraph 3.4.1.1 Coaxial contacts.

Delete first sentence.

Paragraph 3.4.2 Grommet sealing plugs (type II).

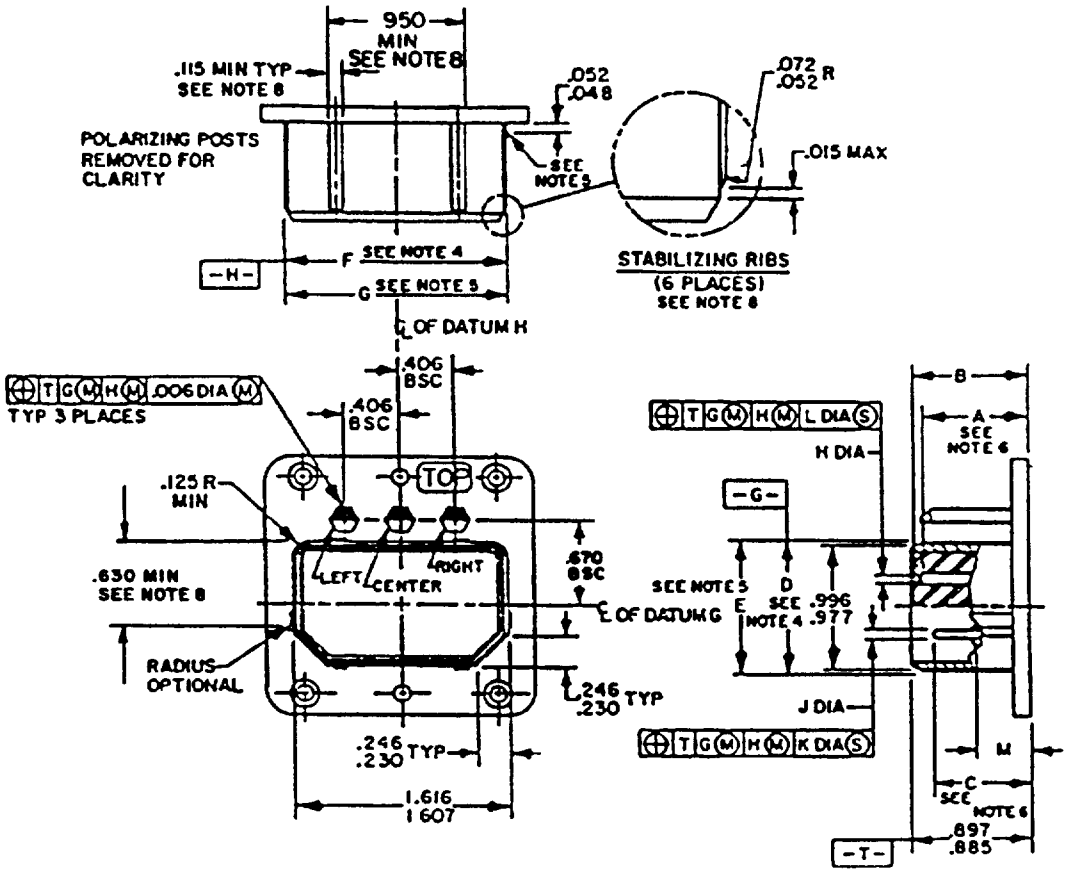
Delete "or MS27187" in first sentence.

Paragraph 3.4.3 Installing and removal tools.

Paragraph should read: "Individual contacts shall be installed with the applicable installing tool as specified in MIL-I-81969/17 and capable of being removed with the applicable removal tool conforming to MIL-I-81969/19 for series 1 connectors. For series 2 connectors, the applicable installing and removal tool conforming to MIL-I-81969/1 or MIL-I-81969/14 shall be used."

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Figure 4, delete the current Figure 4 and substitute the following:



INCHES	mm	INCHES	mm	INCHES	mm	INCHES	mm
.006	.15	.102	2.59	.615	15.62	.910	23.11
.015	.38	.115	2.92	.635	16.13	.977	24.82
.048	1.22	.125	3.18	.670	17.02	.996	25.30
.052	1.32	.230	5.84	.885	22.48	1.607	40.82
.072	1.83	.246	6.25	.890	22.61	1.616	41.05
.098	2.49	.406	10.31	.897	22.78		

FIGURE 4. Single plug connector intermateability control dimensions. (Shell designator A)

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Table, Column D, insert "MIN" and delete dimension ".996 (25.30)."

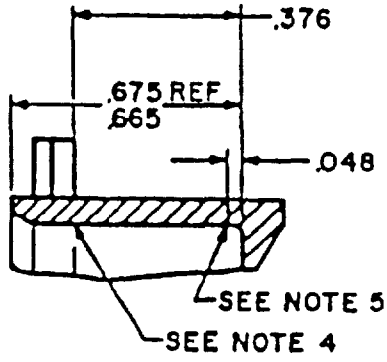
Table, Column F, insert "MIN" and delete dimension "1.616 (41.05)."

Note 4, delete ".102/.098 from front edge"; insert ".654 from flange."

Note 5, delete "/.048."

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FIGURE 5, dimensional changes in upper view as shown:



DETAIL A

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Table, column D, insert "MIN" and delete dimension "1.004 (25.50)."

Table, column E, insert "MAX" and delete dimension "1.003 (25.48)."

Table, column F, insert "MIN" and delete dimension "1.624 (41.25)."

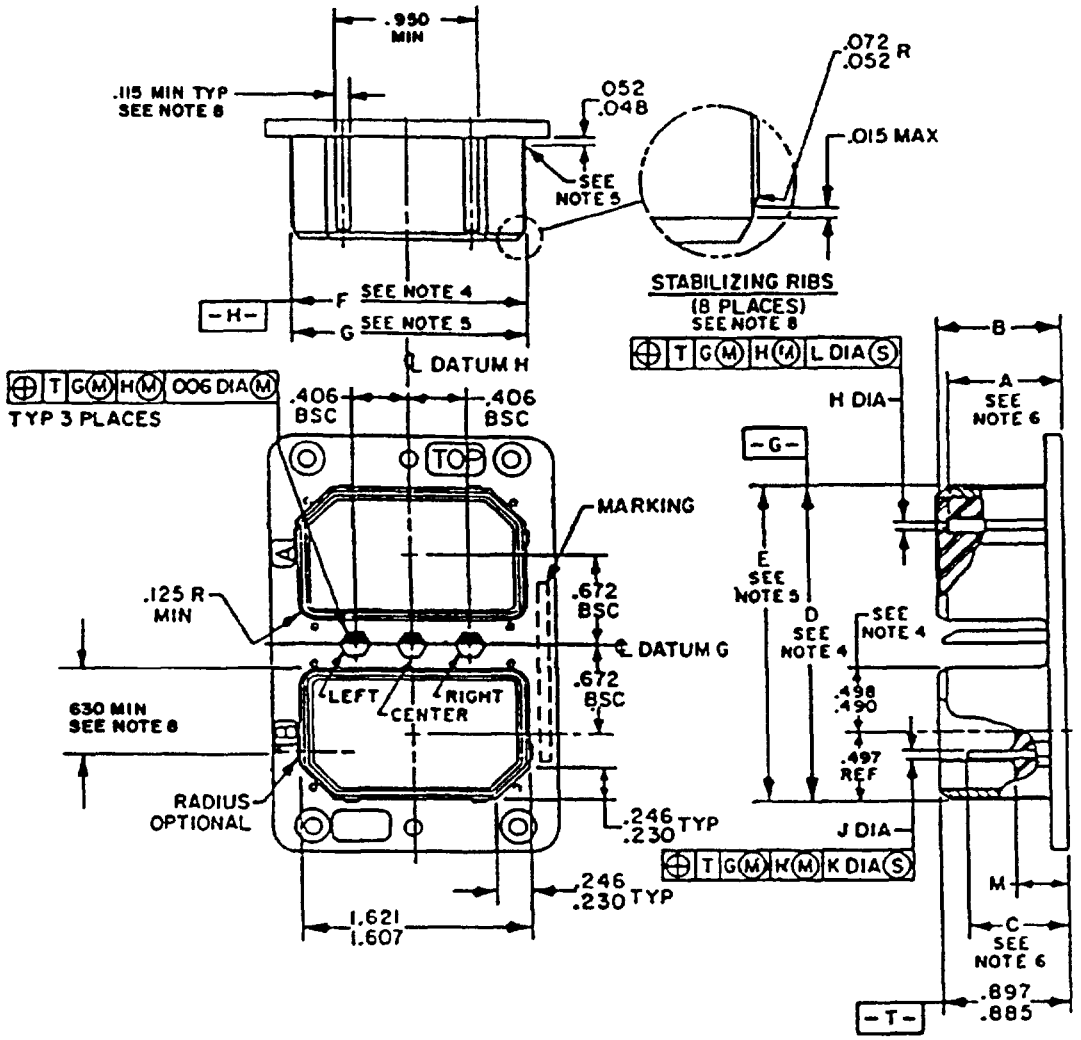
Table, column G, insert "MAX" and delete dimension "1.623 (41.22)."

Note 4, delete ".302/.298 from top edge"; insert ".376 from bottom of cavity."

Note 5, delete ".052/."

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Figure 6, delete current Figure 6 and substitute the following:



INCHES	mm	INCHES	mm	INCHES	mm	INCHES	mm
.006	.15	.102	2.59	.490	12.45	.885	22.48
.015	.38	.115	2.92	.497	12.62	.890	22.61
.048	1.22	.125	3.18	.498	12.65	.897	22.78
.052	1.32	.230	5.84	.615	15.62	.910	23.11
.072	1.83	.246	6.25	.635	16.13	1.607	40.82
.098	2.49	.406	10.31	.672	17.07	1.621	41.17

FIGURE 6. Dual plug connector intermateability control dimensions. (Shell designator A)

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Table, column D, insert "MIN" and delete dimension "2.339 (59.41)."

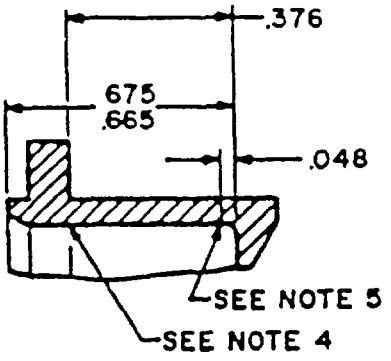
Table, column F, insert "MIN" and delete dimension "1.621 (41.17)."

Note 4, delete ".102/.098 from top edge"; insert ".654 from flange."

Note 5, delete "/.048."

PAGE 23

FIGURE 1, dimensional changes in upper view as shown:



DETAIL A

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Table, column D, insert "MIN" and delete dimension "2.347 (59.61)."

Table, column E, insert "MAX" and delete dimension "2.346 (59.59)."

Table, column F, insert "MIN" and delete dimension "1.629 (41.38)."

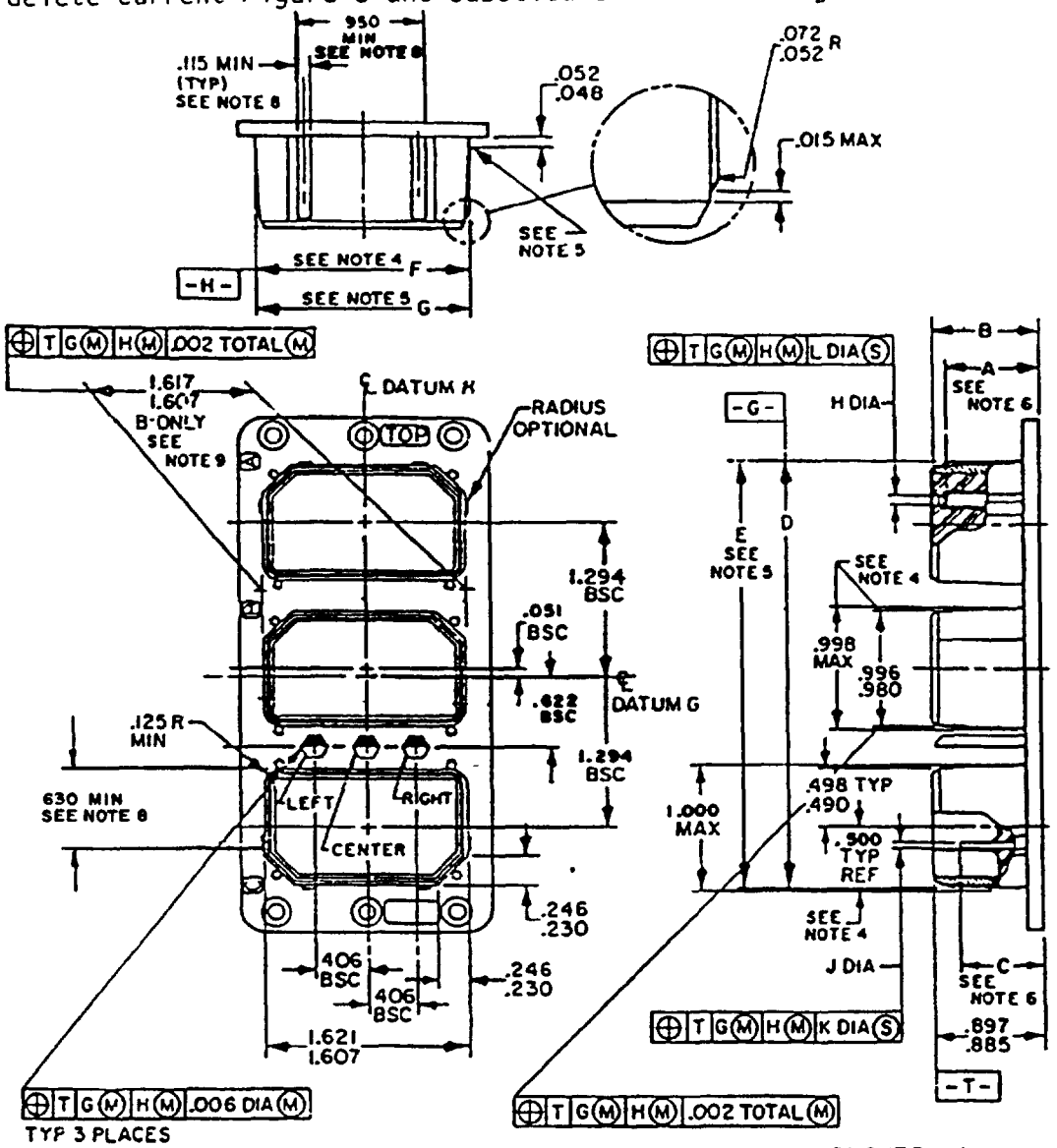
Table, column G, insert "MAX" and delete dimension "1.628 (41.35)."

Note 4, delete ".302/.298 from top edge"; insert ".376 from bottom of cavity."

Note 5, delete ".052/."

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Figure 8. delete current Figure 8 and substitute the following:



INCHES	mm	INCHES	mm	INCHES	mm	INCHES	mm
.002	.05	.102	2.59	.500	12.70	.980	24.89
.006	.15	.115	2.92	.615	15.62	.996	25.30
.015	.38	.125	3.18	.622	15.80	.998	25.35
.048	1.22	.230	5.84	.635	16.13	1.000	25.40
.051	1.30	.246	6.25	.885	22.48	1.294	32.87
.052	1.32	.406	10.31	.890	22.61	1.607	40.82
.072	1.83	.490	12.45	.897	22.78	1.617	41.07
.098	2.49	.498	12.65	.910	23.11	1.621	41.17

FIGURE 8. Triple plug connector intermateability control dimensions.
(Shell designator A)

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Table, column D, insert "MIN" and delete dimension "3.589 (91.16)."

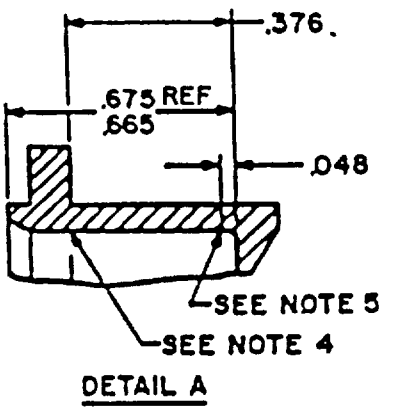
Table, column F, insert "MIN" and delete dimension "1.621 (41.17)."

Note 4, delete ".102/.098 from top edge"; insert ".654 from flange."

Note 5, delete "/.048."

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Figure 9, dimensional changes in upper view as shown:



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Table, column D, insert "MIN" and delete dimension "3.595 (91.31)."

Table, column E, insert "MAX" and delete dimension "3.590 (91.19)."

Table, column F, insert "MIN" and delete dimension "1.629 (41.38)."

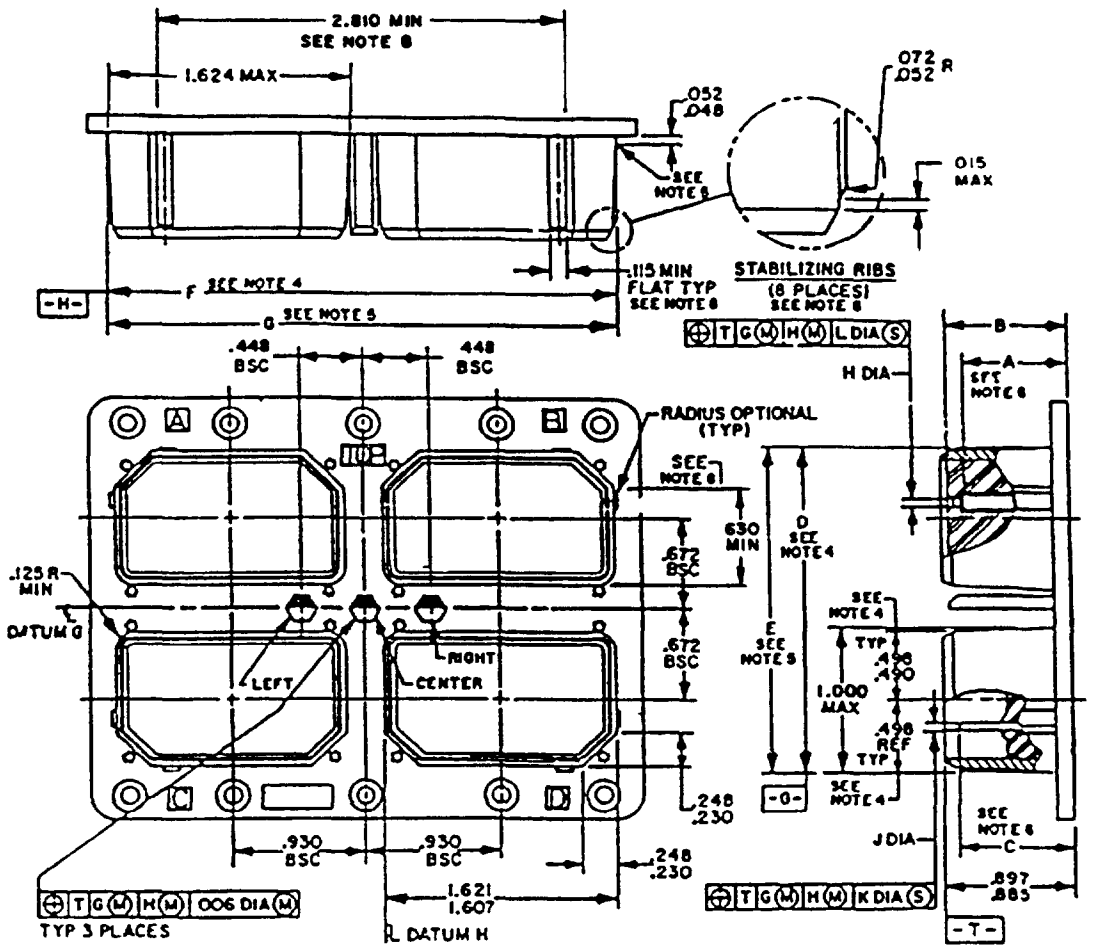
Table, column G, insert "MAX" and delete dimension "1.628 (41.35)."

Note 4, delete ".302/.298 from top edge"; insert ".376 from bottom of cavity."

Note 5, delete ".052/."

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Figure 10, delete the current Figure 10 and substitute the following:



INCHES	mm	INCHES	mm	INCHES	mm	INCHES	mm
.006	.15	.115	2.92	.615	15.62	1.000	25.4
.015	.38	.125	3.18	.635	16.13	1.607	40.82
.048	1.22	.230	5.84	.672	17.07	1.621	41.17
.052	1.32	.248	6.30	.885	22.48	1.624	41.25
.072	1.83	.448	11.38	.897	22.78	2.750	69.85
.098	2.49	.490	12.45	.930	23.62	2.770	70.36
.102	2.59	.498	12.65				

FIGURE 10. Quadruple plug connector intermateability control dimensions. (Shell designator A)

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Table, column D, insert "MIN" and delete dimension "2.345 (59.56)."

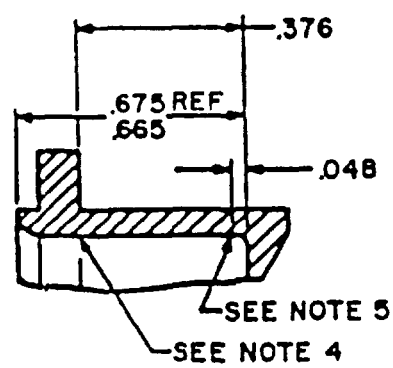
Table, column F, insert "MIN" and delete dimension "3.492 (88.70)."

Note 4, delete ".102/.098 from top of edge"; insert ".654 from flange."

Note 5, delete "/.048."

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Figure 11, dimensional changes in upper view as shown:



DETAIL A

PAGE 32

Table, column D, insert "MIN" and delete dimension "2.351 (59.72)."

Table, column E, insert "MAX" and delete dimension "2.350 (59.69)."

Table, column F, insert "MIN" and delete dimension "3.498 (88.85)."

Table, column G, insert "MAX" and delete dimension "3.497 (88.82)."

Note 4, delete ".302/.298 from top edge"; insert ".376 from bottom of cavity."

Note 5, delete ".052/."

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Army - CR
Navy - AS
Air Force - 85

Review Activities:
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Air Force - 11, 99
DSA - ES

Preparing Activity:
Navy - AS

Project No. 5935-8748

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